Udhayakumar M

03.udhayakumar@gmail.com 9345332063 Rajapalayam, Tamilnadu

https://www.linkedin.com/in/udhayakumar-m-808247209/ github.com/UdhayaDev

I'm a B.Tech Computer Science graduate with hands-on experience in **AI, Data Analytics, and Full-Stack Development**. I work with **Python, SQL, Pandas, NumPy, Excel, Power BI, Tableau, Node.js, React, and LLMs** to build intelligent solutions and perform data-driven analysis. My expertise includes **machine learning, NLP, computer vision, and full-stack web development**. I am passionate about leveraging **data and AI** to solve real-world problems and continuously enhance my technical skills.

EDUCATION

B.Tech(CSE), Karunya Institute of Technology and Science, *CGPA: 7.16* 10/2020 – 05/2024

HSC, Mount Zion Matriculation Higher Secondary School, *68.5%* 06/2019 – 03/2020

SSLC, Mount Zion Matriculation Higher Secondary School, *78.8%* 06/2017 – 03/2018

SKILLS

Programming & Scripting: Python, C, JavaScript, SQL, Node.js, React

Analytics & Visualization: Power BI, Tableau, Excel

AI & Machine Learning: Machine Learning, Data Science, NLP, Computer Vision, LLMs (LLaMA 4)

Web & FullStack Development: HTML, CSS, JavaScript, Express, Socket.io, MongoDB

Tools and Libraries: Groq SDK, VS Code, Postman, Git/GitHub

EXPERIENCE

Data Analyst Intern, *Besant Technologies, Chennai* 07/2025 – 12/2025

- Built efficient Python scripts using Pandas and NumPy to clean, transform, and process large datasets with a focus on performance and readable code.
- Designed and optimized SQL queries with joins and aggregations to support fast and reliable data retrieval for reporting workflows.
- Supported end-to-end reporting features by validating data and ensuring accuracy for real-time use cases such as sales analysis and customer segmentation.

Data Science Intern, *Exposys Data Labs, Bangalore* 12/2023 – 01/2024

- Implemented customer segmentation using K-Means clustering by developing Python logic for data preprocessing and algorithm execution.
- Built reusable preprocessing components for feature extraction, normalization, and validation to efficiently handle structured datasets.
- Evaluated and validated clustering results through analysis and visualization to ensure correctness and reliability of the solution.

Industrial Trainee, *Intel Corporation, Bangalore* 05/2023 – 07/2023

- Implemented a regression-based mobile price prediction solution by developing Python logic for data preprocessing, model training, and evaluation.
- Organized model code into modular, reusable components to improve maintainability and simplify testing.
- Worked with team members to support integration of the trained model into internal forecasting and analysis workflows.

PROJECTS

Sales Analytics Reporting System, *Power BI, Tableau, Python, SQL, Excel*

- Developed a data-driven reporting system using optimized SQL queries and backend data processing logic to support real-time sales and customer analysis.
- Built dynamic KPIs and reporting workflows with efficient data modeling and automated refresh to ensure consistent performance.
- Improved report responsiveness and data accuracy through systematic data cleaning, validation, and calculation optimization.

AI LLM Chatbot , *Node.js, Express, Groq LLaMA 4, HTML/CSS/JS*

- Developed an autonomous AI chatbot powered by LLaMA 4 that responds to user queries with professional, plain-text answers.
- Built a clean and responsive chat UI with a scrollable conversation area and fixed heading, enabling seamless interaction.
- Implemented a lightweight Node.js backend using Express to securely handle AI requests and deliver real-time conversational responses.

Tic Tac Toe -- Full Stack , *React, Node.js, Express, Socket.io, MongoDB*

- Built a real-time multiplayer Tic Tac Toe game with full-stack architecture.
- Implemented live game updates using Socket.io and managed game state with Node.js backend.
- Designed a responsive React frontend and persistent game data storage in MongoDB.

Hand Gesture Recognition & Sentence Generation System, *Python, TensorFlow, OpenCV, LSTM*

- Developed a real-time hand gesture recognition application using deep learning models, achieving approximately 92% classification accuracy.
- Built efficient data preprocessing and model inference pipelines to ensure reliable and low-latency predictions.
- Integrated an LSTM-based text generation module to convert recognized gestures into contextual sentences for assistive communication.

CERTIFICATES

Google Data Analytics Professional Certificate — Coursera • Artificial Intelligence Fundamentals — IBM • Data Visualization — Kaggle • Data Mining — NPTEL • Introduction to IoT & Cybersecurity — Cisco

PUBLICATIONS

Investigation of Hand Gesture Recognition and Script Generation Models

Explored **real-time gesture recognition** and **contextual sentence generation using deep learning** for assistive communication technologies.